

Sean Luka Girgis

Observability / APM Platform Engineer | Telemetry, Performance, Python & Reliability

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Professional Summary

Senior observability, APM, telemetry, and performance engineering professional with 20+ years of enterprise experience across Dynatrace AppMon/Synthetics, CA APM/Introscope, AppDynamics, BMC TrueSight/TSCO, telemetry analysis, dashboards, alerting, production troubleshooting, Python scripting, SQL, cloud data workflows, and system reliability. Strong background supporting large-scale monitoring environments, transforming infrastructure and application telemetry into actionable reliability insights, defining monitoring standards, troubleshooting production issues, and documenting support/runbook practices. Best aligned to observability platform roles requiring telemetry pipelines, metrics/logs/traces concepts, APM platforms, dashboarding, alerting, performance engineering, Python automation, cloud exposure, and production reliability; Prometheus, Grafana, OpenTelemetry, Kubernetes, GPU-cluster observability, and IaC/deployment automation are positioned carefully as adjacent or learning areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Supported enterprise observability, capacity, and production reliability workflows across BMC TrueSight/TSCO, CA Wily/APM, AppDynamics, Oracle, AWS-backed reporting layers, and large-scale infrastructure telemetry feeds.
- Built Python/Pandas/PySpark and SQL workflows that processed telemetry from 6,000+ infrastructure endpoints into validated dashboards, reports, forecasting datasets, and operational insights.
- Analyzed metrics, utilization trends, latency symptoms, missing feeds, abnormal deltas, and reporting discrepancies to help teams identify reliability risk and capacity bottlenecks.
- Developed validation checks for telemetry freshness, row counts, missing data, unmatched records, abnormal metric movement, and output consistency to improve trust in monitoring and reporting data.
- Troubleshoot production issues by tracing source feeds, metric behavior, Python processing, SQL transformations, failed outputs, latency symptoms, and downstream reporting discrepancies.
- Created runbooks, metric definitions, dashboard notes, validation checkpoints, support procedures, and troubleshooting documentation to improve repeatable operational response.
- Partnered with infrastructure, application, analytics, engineering, and business teams to explain telemetry findings, coordinate fixes, and improve production reliability.
- Used AWS S3, Glue, Athena, Redshift, EC2/ECS concepts, Python, SQL, and monitoring data to support cloud-oriented observability and reporting patterns.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed Dynatrace AppMon and Gomez Synthetic Monitoring for Brand.com and critical hospitality systems, improving visibility into user-facing reliability and performance.
- Correlated real-user monitoring, synthetic transaction data, and load-test behavior to identify bottlenecks, user-impacting symptoms, and improvement opportunities.

- Led the FAST project, mining real-user and synthetic monitoring telemetry to surface optimization recommendations adopted by engineering teams.
- Created dashboards and reporting views that converted raw monitoring signals into actionable performance, reliability, and stakeholder insights.
- Supported AWS cloud migration planning by evaluating monitoring readiness, transaction visibility, and performance reporting needs.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as CA APM/Introscope SME for a large financial-services environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Defined monitoring standards, dashboards, alert rules, SLA thresholds, validation checks, and operational reporting artifacts for application, middleware, infrastructure, and operations teams.
- Provided production troubleshooting guidance using CA APM telemetry, transaction visibility, dashboards, alerts, J2EE/WebLogic monitoring, and performance-analysis patterns.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry, replacing manual reporting workflows with repeatable observability processes.
- Delivered onboarding support, training material, technical documentation, and knowledge sharing to improve adoption of monitoring and troubleshooting practices.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed J2EE telecom applications under load to identify throughput limits, JDBC bottlenecks, thread contention, heap pressure, CPU constraints, memory issues, and garbage-collection symptoms.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve runtime visibility during performance and production-readiness testing.
- Documented performance baselines, defects, risks, telemetry findings, and remediation recommendations for engineering and release teams.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led validation and production-readiness support for a high-throughput shopping engine migration from 200+ MySQL nodes to a 6-node Oracle RAC cluster.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.

Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC) (2007-10 – 2008-05)

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, VC++ components, and DB2-backed integration modules in a government modernization environment.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (2001 – 2007)

- Developed and supported billing, customer-management, and operational interfaces using C, C++, Pro*C, Oracle, PL/SQL, SQL scripts, XML, Java/J2EE concepts, and enterprise messaging patterns.

- Built automation scripts in KornShell/ksh, Perl, Awk, Syncsort, and SQL to support extraction, transformation, reporting, deployments, validation, and production housekeeping.
- Provided production support, defect investigation, troubleshooting, issue escalation, release support, and QA-cycle coordination for telecom billing and customer-management applications.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science

Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering

Zagazig University, Egypt

Skills

Flagship Projects

Enterprise APM Reporting Automation

Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry across large financial-services monitoring environments.

Technologies: CA APM, KornShell/ksh, Perl, SQL, Unix/Linux, Automation

FAST Performance Analytics Project

Mined Dynatrace and Gomez Synthetic Monitoring telemetry to identify optimization opportunities, improve transaction visibility, and communicate performance recommendations to engineering teams.

Technologies: Dynatrace AppMon, Gomez Synthetic Monitoring, Performance Analytics, Dashboards, APM

Production Telemetry Quality and RCA Workflow

Built validation and troubleshooting workflows for telemetry-backed reporting datasets, including freshness checks, missing feed diagnosis, abnormal trend detection, prior-run comparisons, and runbook documentation.

Technologies: Python, SQL, Pandas, BMC TrueSight, AppDynamics, RCA, Runbooks

ServiceCall AI RAG Demo

Built staged Python proof-of-concepts with FastAPI service layers, Docker-first smoke tests, Pydantic contracts, deterministic endpoints, and testable service behavior.

Technologies: Python, FastAPI, Docker, Pydantic, pytest, RAG